

Trier University
Faculty VI - Regional and Environmental Sciences
Department of Environmental Remote Sensing and Geoinformatics

**POTENTIAL CONFLICT BETWEEN THE CULTURAL LANDSCAPE
AND ANIMALS: A CASE STUDY OF THE RUWER VINEYARDS
AND WILD BOARS**

Term Paper for
ADVANCED METHODS IN GIS AND APPLICATIONS TERM PAPER

Summer Term 2024
Lecturer: *Melanie BRAUCHLER*

Felix Zimmermann 1675850 s6fezimm@uni-trier.de	Karel Loic NIMPA FOKOUA 1734880 <u>s6kanimp@uni-trier.de</u>	Jan-Niklas Seitz 1417149 s6jaseit@uni-trier.de
--	--	--

Trier, 09/10/2024

Declaration of Authenticity

I hereby declare that I have written the present work independently, without assistance from external parties and use of other resources than those indicated. The ideas taken directly or indirectly from external sources (including electronic sources) are duly acknowledged in the text. The material, either in full or in part, has not been previously submitted for grading at this or any other academic institution.

I understand that if at any time it is shown that I have significantly misrepresented material presented to Trier University, any degree or credits awarded to me on the basis of that material may be revoked.

Trier, 10/10/2024

City, Date

J. Seitz Zimmermann

Signature

Consent to Digital Plagiarism-Detection Software

In accordance with the provided Declaration of Authenticity, I consent to having my work checked with digital plagiarism-detection software.

Trier, 10/10/2024

City, Date

J. Seitz Zimmermann

Signature

The **Declaration of Authenticity** and **Consent to Digital Plagiarism-Detection Software** directly follow the **Title Page** or can be provided separately.

Table of content

1. Introduction	1
2. Study Area and Datasets	1
2.1. Study area and its physical environment	1
2.2. Data acquisition	3
3. Methodology	3
3.1. The potential location of vineyards	3
3.2. Damage risk analysis using Cost distance analysis	5
4. Results	7
4.1. The cost distance analysis	8
5. Discussion	9
5.1. Potential improvements for the identification of suitable vineyard locations	9
5.2. Wild boar and Ruwer vineyards	10
6. Conclusion	11
7. References	12
8. Appendix	15

Table of figures

Figure 1: Overview of the study area	2
Figure 2: Vineyards suitability model	4
Figure 3: Virtual Boar model	5
Figure 4: Entire of the cost distance analysis	7
Figure 5: Vineyards Areas and location	8
Figure 6: Boar location path	9
Figure 7: Solar Analysis result	15
Figure 8: Cost Distance Surface	16

1. Introduction

Today, as the world's population increases, habitat loss and fragmentation are identified as the most significant current threats to biodiversity (IPBES, 2019). At the same time, habitat degradation reduces the quality and availability of food for wildlife in natural systems and intensifies potential conflicts between crops and wildlife (Li et al., 2023). These conflicts can lead to losses in the livelihoods of local populations. In Europe, wild boars (*Sus scrofa*) and agricultural landscapes often have a conflictual relationship, wild boar numbers in many countries have increased over the last three decades (Pittiglio et al., 2018). The wild boar's diet consists mainly of plants; thus, an abundant food source that appears locally is often exploited, and conflicts with agricultural areas have resulted from this behaviour (Schley et al., 2008). In Europe, these conflicts are mainly due to trampling, consuming crops, and rooting in grasslands. This situation is now observed in Ruwer Valley.

The Ruwer Valley is part of Moselle wine-growing region and is known for its high-quality wines. Due to its proximity to forests, towns, and rivers, the cultural landscape of the Ruwer vineyards reflects a shared landscape between man and nature. This landscape is also home to various wild animals, particularly wild boar, which increasingly conflict with vineyards. In Europe, data from recent hunting statistics suggest that the number of wild boars has been steadily increasing since the 1980s (Massei et al., 2015). Mapping of wild boar density in Europe reveals a high density of wild boar in western Germany (Pittiglio et al., 2018). This also refers to an increase in the damage caused by wild boars to vineyards, which has led to the abandonment of certain plantations by wine growers (Bruns, 2021).

In the research for solutions to human-wildlife conflict, GIS has often been used to spatially map the geographical distribution, identify populations at risk, and examine potential drivers of conflict. This term paper aims to understand the potential conflict between Ruwer-valley vineyards and wild boars using Geoinformatics methods. Two specific topics will be analysed. First, the potential location of vineyards using environmental pre-request and, second, the risk level of wild boar damage in the identified vineyards using cost distance analysis. The result of this analysis will help to validate the hypothesis that areas with abandoned vineyards are no longer in use due to increased risk of wild boar damage.

2. Study Area and Datasets

2.1. Study area and its physical environment

The chosen area for this study is the Ruwer Valley wine-growing region, an affluent of the Moselle River near Waldrach in western Germany. It's located in the state of Rhineland-Palatinate (Fig. 1). The study area covers an area of about 28 680 ha, with a landscape made up of a combination of different types of land use: forest, various vineyards, urban areas, and water. It is part of the Rhenish slate mountains. The Ruwer valley descends from a plateau formed by the Hunsrück, from around 500 m altitude in the south to around 200 m altitude in the north (Richter, 1980).

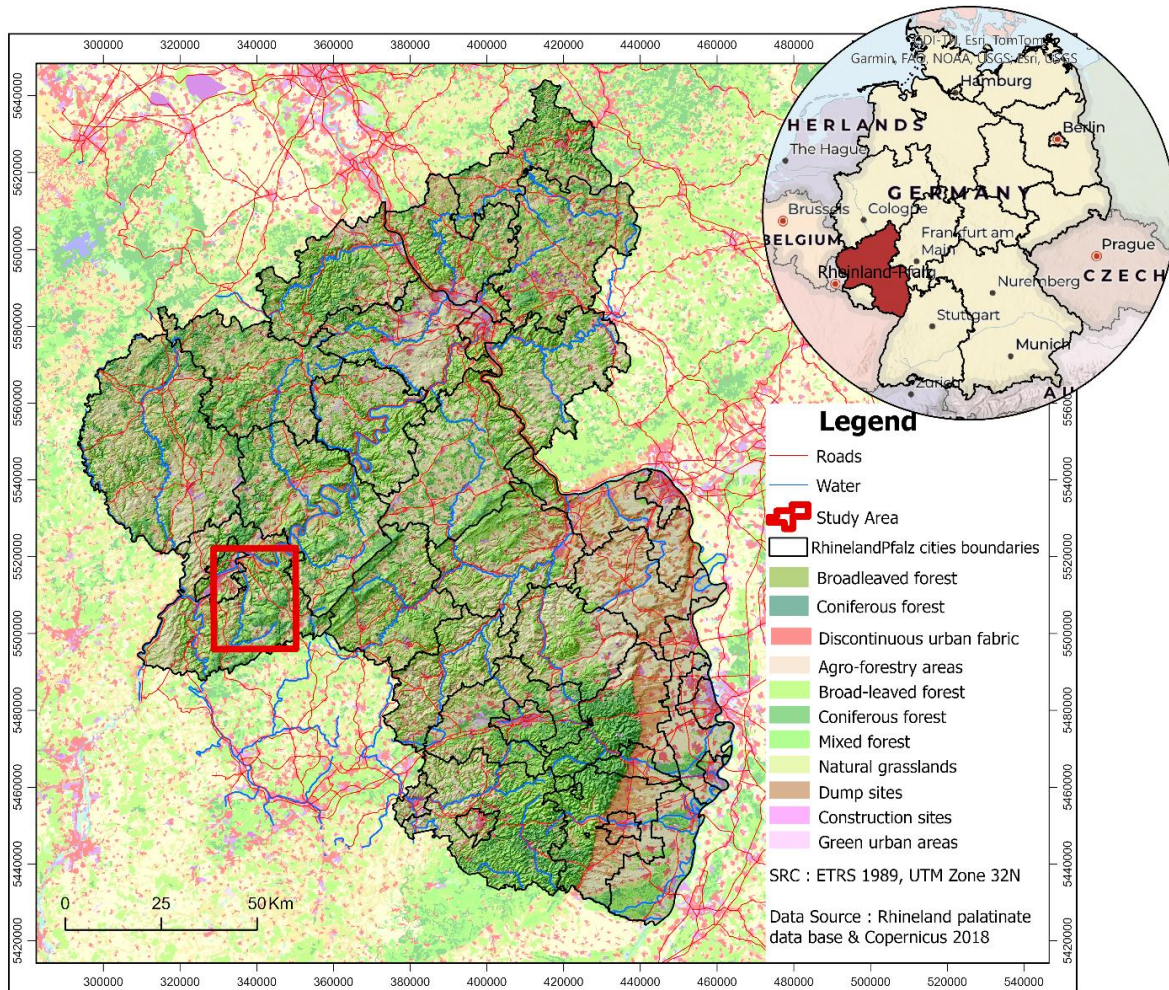


Figure 1: Overview of the study area

The average annual rainfall of the study area is 765 mm, with a maximum of 65-72 mm per month during summer. The lowest monthly rainfall is recorded between February and April, with a value between 50 and 60 mm per month. The average annual temperature is 9.3°C, with highs of 16.2-17.6°C in June, July, and August, and lows of 1.5-2.3°C in January and December (Rodrigo-Comino et al., 2017).

The soil of the Ruwer valley is a mixture of mineral and organic matter, organisms, water and air. The mineral components are formed by the impairment of the rock subjacent mother. Typical rocks and soils consist of non-calcareous loamy slate soils (Devonian slate), non-calcareous loamy sandstone soils, sand and gravel soils, calcareous loam and clay soils on marl and dolomite (Ministerium für Wirtschaft Klimaschutz, Energie und Landesplanung Rheinland-pfalz & Ministerium für Umwelt, Landwirtschaft Ernährung, Weinbau und Forsten Rheinland-pfalz, 2015).

The main rivers in the study area are the Ruwer river and its affluent. The Ruwer is a right affluent of Moselle, which flows south from a plateau on the Hunsrück mountains, descending from an altitude

of around 500 m to around 200 m in the north (Richter, 1980). According to the Natura 2000 classification, several habitats are present in the Ruwer Valley, including forests and herbaceous meadows; regarding the fauna, there are several species of fish, beetles, butterflies, and mammals such as bats and wild boars (Landesamt für Umwelt Rheinland-Pfalz, 2017).

2.2.Data acquisition

Two types of data were used in this analysis. Firstly, secondary data from work on the physical characteristics of vineyards and the behaviour of wild boar. Secondly, the study exploited the primary data collected through the database, observations, and field data. To identify the potential locations of vineyards using environmental pre-request, we used a 25 m resolution Digital Elevation Model (DEM) of the study area. Additionally, a range of aspect and slope values for wine growing were derived from literature. Corine land cover database and land used field data were used to assess the risk level of wild boar damage. A digital orthophoto at a resolution of 20cm from the Geoportal of Rhineland-Palatinate was used for further editing of the Corine land cover database.

3. Methodology

The methodology used in this analysis varied according to the research objectives.

3.1.The potential location of vineyards

After acquiring the digital elevation model of the study area, the data was reprojected into the local coordinate system, which is ETRS 1998 UTM zone 32N. Following that, crucial criteria of vineyards were identified in literature and applied to our context: slope, orientation, and altitude. Because of soil erosion, investigations have shown that the slope in the Ruwer Mosel Valley is between 0 – 50°, corresponding to a maximum of 119% (German Wine Institut, 2024). As a slope of more than 15% makes vineyard management difficult for vine growing, it is preferable to have a slope between 0 and 15%. In addition, tractors and other machinery can have difficulty working on steep slopes (Grape to Glass, 2024). However, to take all these factors into account, as well as erosion reducing vineyard areas, wine growers can produce without taking account gentle or medium slopes, so the slope threshold has been set between 10 and 25% for this analysis.

Because of sun exposure, aspect is also essential for the vine's growth (photosynthesis and ripening). However, excessive exposition to the sun can also be harmful to the vines. South-facing vines receive the most sunlight, which is why an orientation from South-east to South-west was chosen to reduce the amount of heat reaching the plants. Because of low temperatures at higher elevations and to avoid the risk of night frost, no altitude above 260 m was permitted (Neuner et al., 2020).

Using all these parameters, the simple vineyard suitability model was created using the ArcGIS Pro model builder. Following this step, the watershed was calculated from the DEM. As wild boars will not restrict themselves on watershed borders, a buffer of 3 km was created. Then the Solar radiation (diffuse radiation, and direct duration) was computed for the entire study area for the period of March to October which corresponds to the growing season of vineyards. Afterwards, the mean of these data sets was calculated using a zonal statistics tool. According to Wosc hek (2014) vineyards require sun exposure between 1800 and 2000 hours. Considering this threshold, new suitable vineyards were

extracted from the first vineyard polygons of the simple suitability model. The final model can be seen in the figure 2.

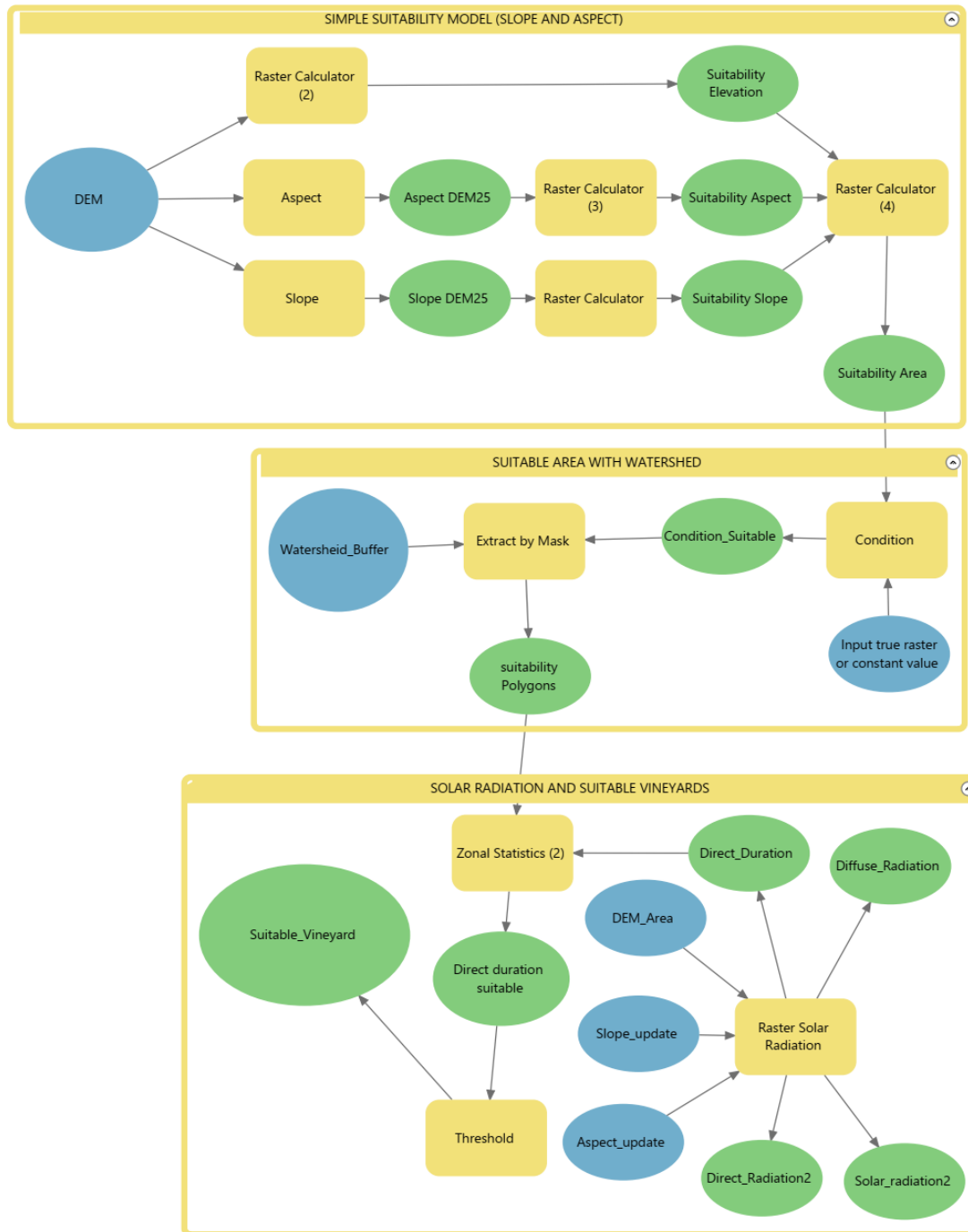


Figure 2: Vineyards suitability model

3.2.Damage risk analysis using Cost distance analysis

To analyse the risk level of wild boar damage in vineyards, the Corine land cover classification was used and updated based on a digital orthophoto of the study area. By means of the orthophoto, which was acquired from the Geoportal of Rhineland-Palatinate (Geoportal RLP, 2024) , the different land cover polygons were edited or further divided.

The workflow for the creation of the virtual boars is shown by figure 3. Therefore, all forest areas were extracted from the updated Corine land cover classification with the function “Dissolve”. Afterwards, the “Buffer” function was used with the buffer distance set to -300, to determine the forest core areas. To be able to calculate the boar number per forest patch, the use of the “Multipart to Single part” function is necessary. Before the last step two fields were added to the attribute table. The first field was added to calculate the area in hectares for each forest patch. In the second field the number of boars for each forest patch were calculated by the area in hectares multiplied with 0.05. This number was chosen, because according to Hohmann (2013) 250 boars are expected to live on 5000 ha, which leads to 0.05 boars per hectare. For an appropriate analysis all patches with a boar amount of 0 were set to 1. Finally, the function “Create Random Points” was used with the core forest areas single parts as constraining feature class and the number of boars as the number of points to create the virtual boars.

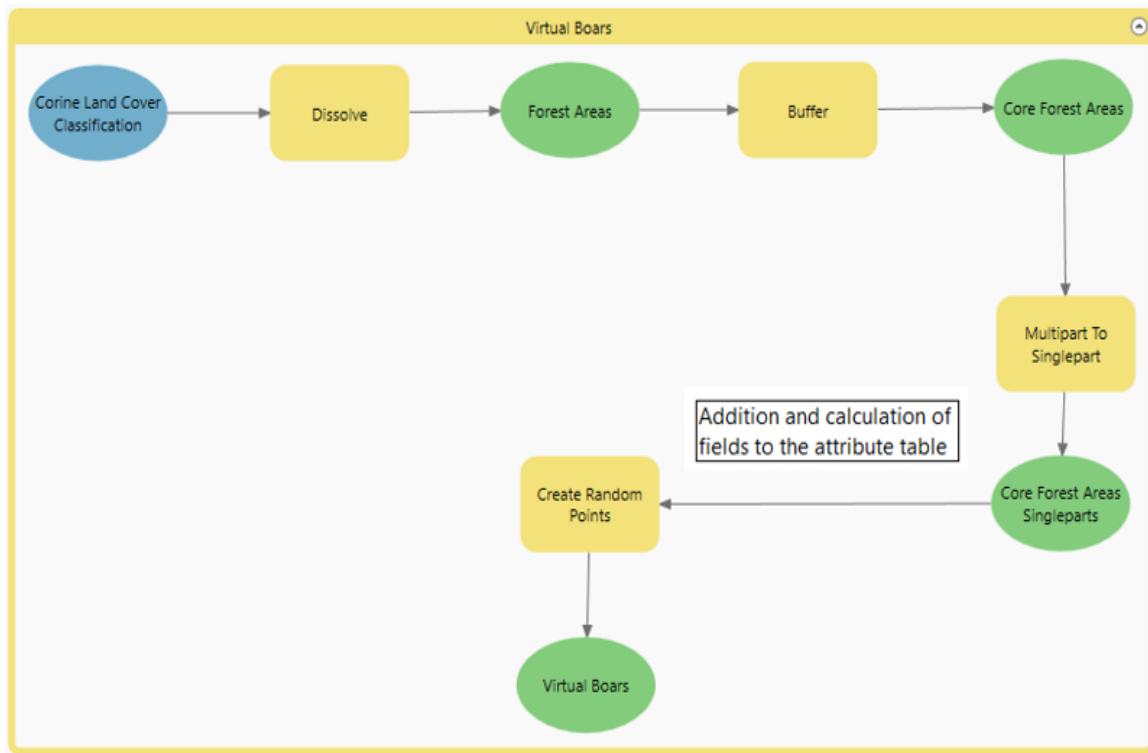


Figure 3: Virtual Boar model

Before continuing with the analysis, the polygons of the suitable vineyards had to be slightly adjusted. The polygons contained in the dataset had sizes between 0.04 and 30 ha, as no size limitations were

defined in the suitability model. However, the smallest polygons were only snippets and therefore not realistically suited for the construction of a vineyard. Since keeping these polygons in the dataset might have a negative impact on the result of the cost distance analysis, all suitable vineyards with a size below 0.5 ha were excluded.

Next, the cost distance analysis was performed in ArcGIS Pro to assess the damage risk of boars to the determined suitable vineyards. This process required the suitable vineyards, virtual boars and a cost surface raster as data inputs. The cost surface raster was created as a combination of three raster datasets, which are the Corine land cover classification raster, the slope raster that was used during the calculation of the suitable vineyards, and the roads and paths in the study area. Roads and paths were considered as additional input for the final cost surface, as boar damage seems to occur in areas that are further away from main roads (Cappa et al., 2019). This is most likely caused by the presence of loud noises, created by high amounts of traffic on main roads. Therefore, the road vector dataset from the OpenStreetMap project was downloaded and transformed into a raster dataset. The different road types were then reclassified on a scale between 1 and 5, depending on how likely it is that boars would cross the specific road type. For example, motorways were classified with the value 5 (boars will most likely not cross these roads, only if they have to), while forest paths were classified with the value 1 (boars will definitely cross these paths).

As a result of an altitude difference of around 300m between the northern and southern parts of the study area, the slope dataset was included in the final cost surface raster. While boars are capable of traversing more difficult terrain, the assumption was made that they rather traverse areas with less steep slopes, as they are easier accessible. Therefore, the dataset was again reclassified on a scale between 1 and 5, where gentle slopes were assigned with the value 1 and very steep slopes with the value 5.

Different land cover types have a high influence on the traversal patterns of boars, which is why the Corine land cover classification was used as a third dataset for the cost surface raster. Boars prefer landscapes with cover when travelling, such as forests and shrub lands. On the contrary, boars tend to avoid urban and other man-made areas with a higher presence of human activities (Cappa et al., 2019). Similar to the roads and slope datasets, the Corine land cover classification was also reclassified on the same scale. Each land cover type was assigned a value between 1 and 5, depending on how likely it is that boars would traverse the specific land cover type.

Before combining all three datasets, each raster was assigned with a different weight based on their importance for the final cost surface raster. The Corine land cover classification received the highest value (0.6), as it has the most influence on the travel path of boars, followed by the slope (0.25) and the roads (0.15). Afterwards, the raster calculator was used to combine the three weighted rasters into the final cost surface raster, which was then used as input, alongside the suitable vineyards and virtual boars, in the “cost distance” and “cost path” tools of ArcGIS Pro. It’s important to mention that the suitable vineyards were used as the path source of the “cost distance” tool. This decision was made to ensure that each point of the virtual boar dataset receives one path to the closest vineyard. The result of the cost distance analysis can be seen in figure 5. Furthermore, the model containing the entire workflow of the cost distance analysis is displayed in figure 4.

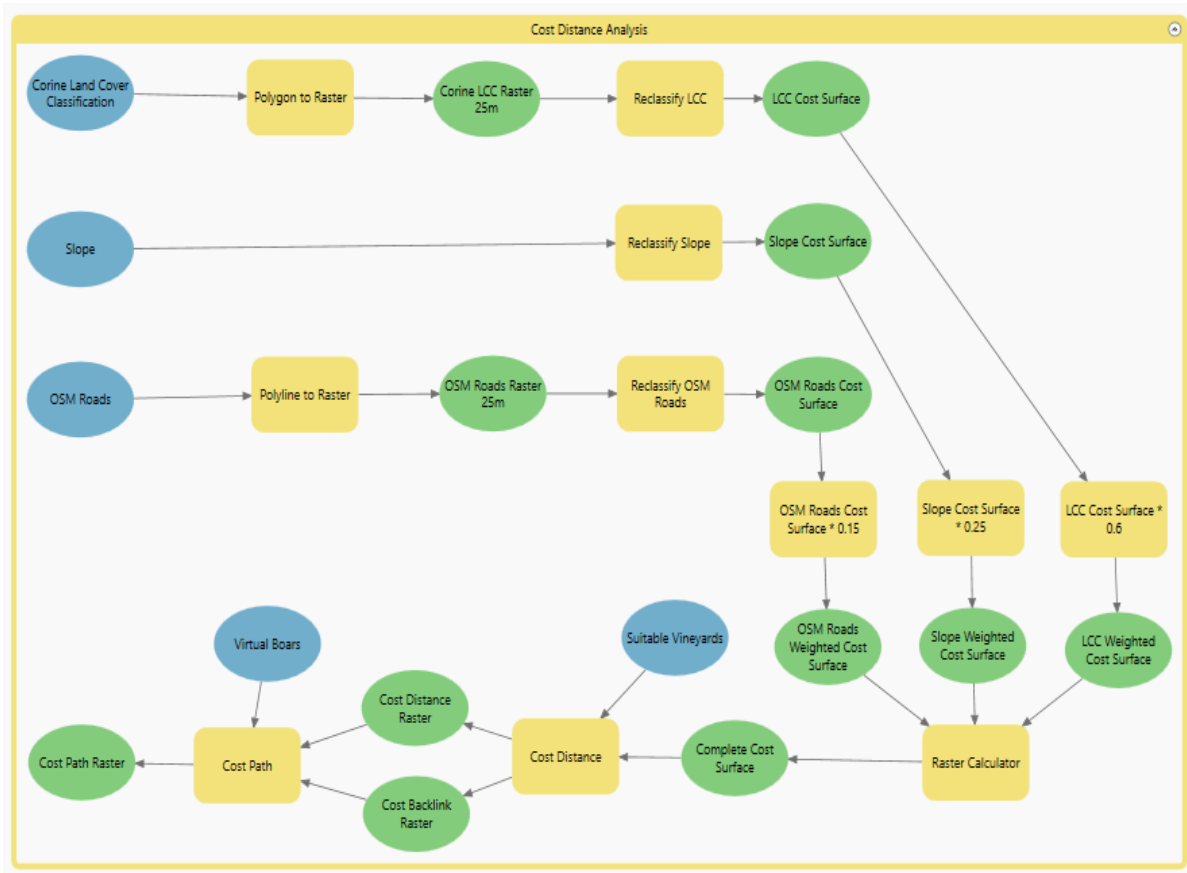


Figure 4: Entire of the cost distance analysis

4. Results

This part of the study presents all the obtained results, visualized in additional maps.

The potential location of vineyards

After the first part of this analysis, the result for the suitable vineyards is shown in figure 5. Most of the areas are located in the northern parts of the study area, especially in the low altitude parts of valleys, as can be seen by the dark blue zones. The centre of the study area is characterised by only a few smaller vineyards, whereas the southern parts of the study area contain no suitable areas for vineyards. These zones correspond to the different characteristics set for the suitable vineyards (slope < 25%, south-east to south-west orientation, with a maximum altitude of 260 m and at least between 1800 and 2000 sun hours). 15 of the identified areas have a size above 10 ha, while more than half of all vineyards are smaller than 2 ha. The largest vineyard covers an area of 30 ha.

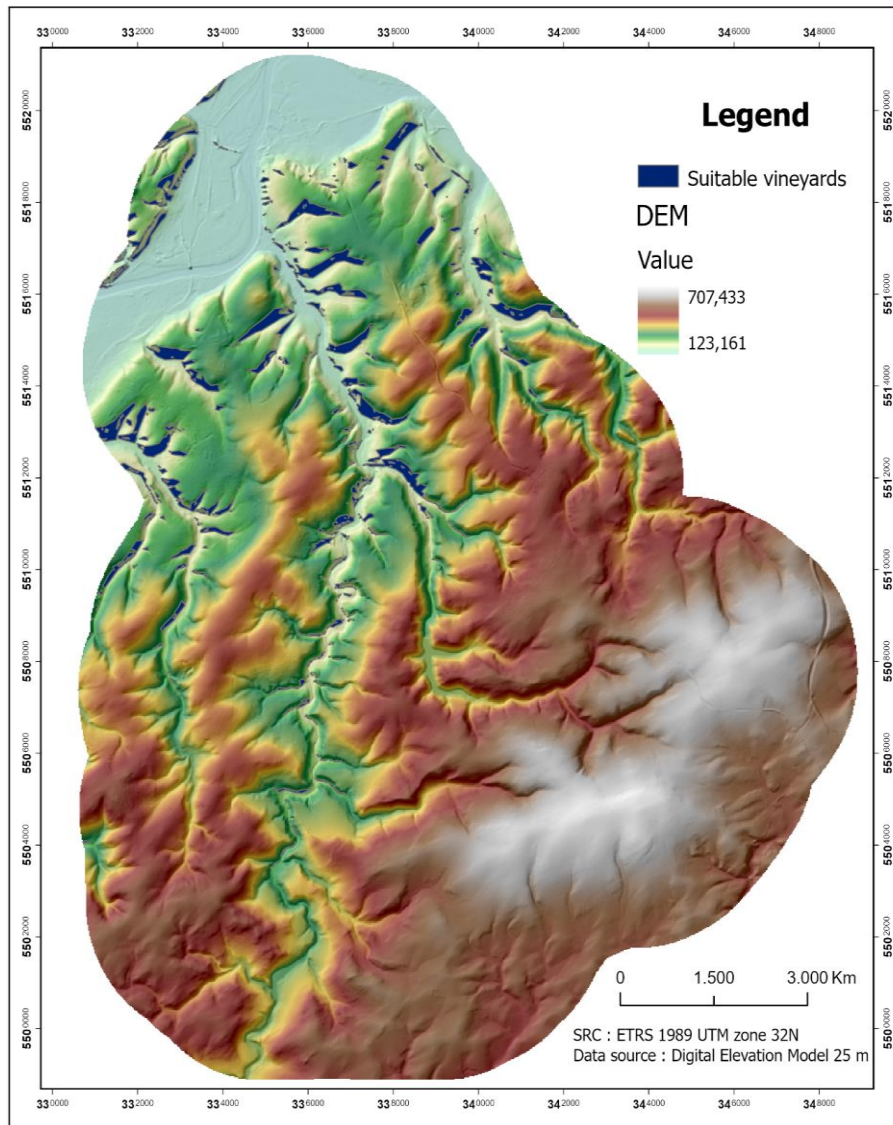


Figure 5: Vineyards Areas and location

4.1. The cost distance analysis

The result of the cost distance analysis is shown by figure 6. The majority of virtual boars live in the southern part of the study area. Therefore, they move especially to the smaller vineyards in the centre. Many boars are in competition for these vineyards. Some of them travel long paths to a vineyard, because of the lack of suitable vineyards in the south. In the middle and south-east of the study area some virtual boars even migrate a long way until they reach small vineyards in the lower parts of the northern area. Only a few virtual boars are in the northern part of the study area. These boars prefer to move to smaller vineyards near their initial location. Consequently, the large vineyards in the northern part of the study area are not much affected by wild boars.

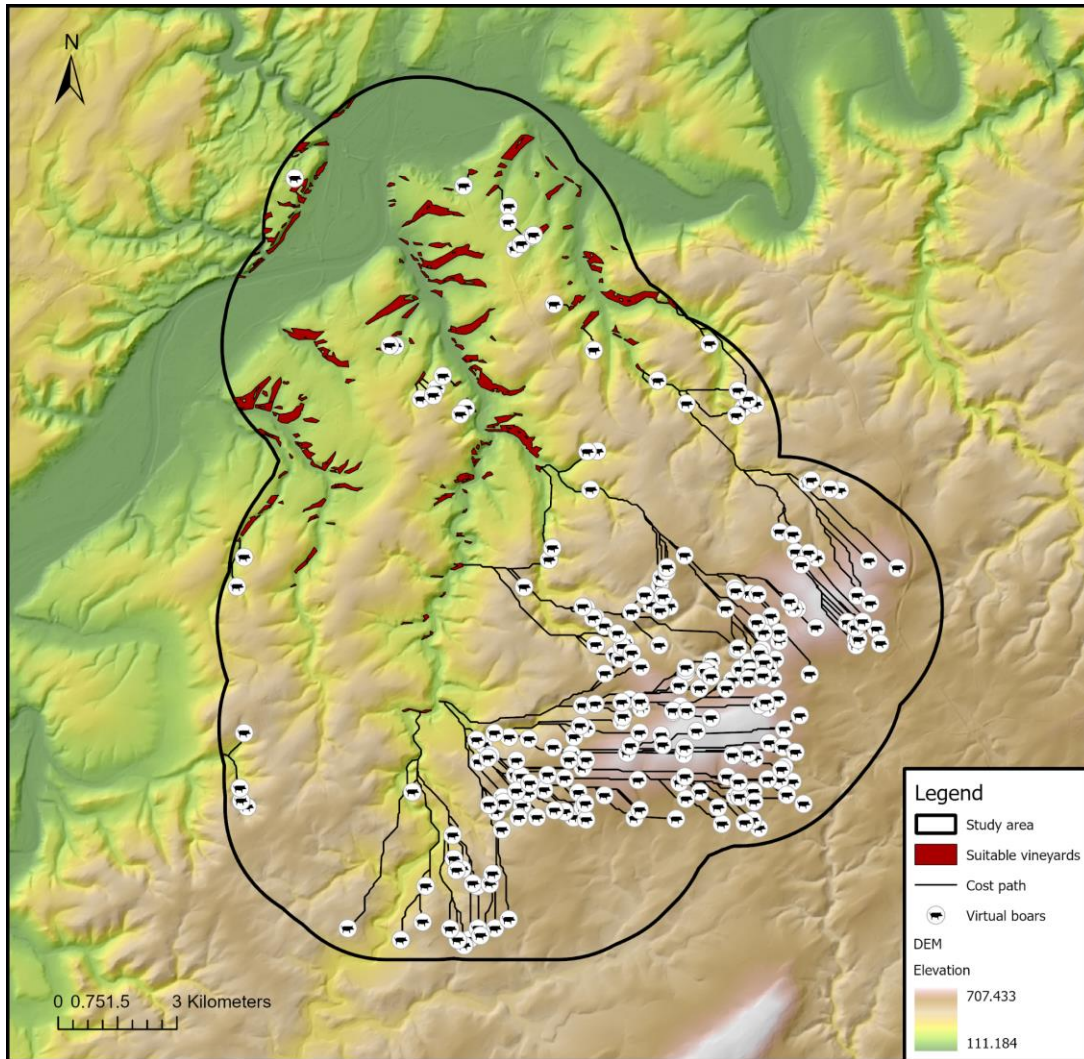


Figure 6: Boar location path

5. Discussion

5.1. Potential improvements for the identification of suitable vineyard locations

The results of the simple suitable vineyards approach seem overall very promising. All identified vineyard locations match the set parameters of the model, as was already described in chapter 3.1. The abundance of suitable vineyard locations in the northern part and the lack of suitable vineyard locations in the southern part of the study area mostly correspond to the altitude and slope parameters. As can be seen in figure 5, the southern regions are largely characterised by higher altitudes. Additionally, the existing slopes are also steeper in comparison to the recommended values.

However, the resulting vineyards contain one major error, which needs to be addressed and corrected. During the calculation process of the vineyard areas, only topography and radiation were considered

as parameters to determine the suitability of an area. While the application of these parameters may be sufficient in non-urban areas, they can result in an overlap between suitable vineyard locations and urban structures, as is the case in the resulting data. Some of the vineyard areas, especially the larger ones, overlap partwise with existing urban areas and structures and should therefore not be identified as suitable locations. This can easily be corrected by adding urban areas as an additional criterion for the identification of suitable vineyard areas, either by using the urban area category from the Corine land cover dataset or by applying buffers to other data containing urban structures.

Furthermore, additional criteria can be used to narrow down the resulting suitable vineyard locations even further. One example for a potential criterion is the soil type, which has a strong influence on the wine quality. Potassium, magnesium, and nitrogen are a few examples of important nutrients that vines need to withdraw from the soil. Therefore, soils that lack the required nutrients should be excluded from the analysis. It is also important that the underlying soils are not too fertile, as an overabundance of nutrients not only leads to an overproduction, but also results in a quality loss of the grapes (DELINAT, n.d.). To include this criterion in the analysis, one or more suitable datasets are required. Available online resources like the soil type maps from the “Landesamt für Geologie und Bergbau” in Rhineland-Palatinate can be used for this task. Additionally, soil samples can be taken during field work in the study area to analyse their properties.

5.2. Wild boar and Ruwer vineyards

The result of the cost distance analysis shows great potential for the installation of vineyards in the northern part of the study area, because of the lack of forest areas in this part. This leads to a lower wild boar distribution. Furthermore, only small vineyards in the northern area are threatened by wild boars. It is noticeable that these wild boars are located near their initial location. From the perspective of the wild boars this behaviour makes sense as they can forage vineyards with a low risk, because they have the shelter of the forest nearby. However, the analysis does not consider the scenario of what happens when the small vineyards do not provide enough food for the wild boars anymore. Then the wild boars could travel further away from the forest to the larger vineyards. This would for example also be the case, if only the larger vineyards would be established, because they are more profitable for vintners. As a result, the competition between the wild boars for the larger vineyards could increase. To cover both scenarios, the exact number of wild boars, their food demand and the yield of vineyards per hectare are required to estimate the duration of food availability. Based on these parameters already established vineyards can also be considered, as they could stop wild boars from further migration, if they offer sufficient food supplies. For these scenarios, new or adjusted cost distance analyses must be conducted.

According to the analysis the vineyards in the middle part of the study area are at high risk of being exploited by wild boars. This can be explained by the large forest areas in the south, which provide a vast habitat for the wild boars. In this part of the study area is a high competition between the wild boars for the few vineyards in the centre. Therefore, it is likely that the vineyards are foraged in a short amount of time, which would lead to some wild boars travelling further in the north eventually reaching the larger vineyards.

In general, the analysis has some weaknesses that must be discussed. First, the consideration of the roads is probably not so meaningful. Wild boars are mainly nocturnal (Brivio et al., 2017), where there is nearly no traffic or loud noises on the streets, which makes it unlikely that wild boars avoid streets on their paths, even highways. So, the weight of the roads in the cost distance analysis should be decreased.

Second, the values of the different land cover classes for the cost distance surface could be optimised with additional literature or with recommendations of experts. However, it is very difficult to predict the exact behaviour of wild boars, because every individual and also different sounders of wild boars show distinct behaviour patterns regarding their willingness to take risk for foraging. Today, some wild boars even invade urban areas e.g. Berlin (Nabu).

Third, hunting impacts the spatial behaviour of wild boars and could be considered in the analysis (Scillitani et al., 2009). Nevertheless, the magnitude of this effect is highly variable and difficult to implement in a GIS cost distance analysis. Therefore, datasets of the hunting zones in the study area are necessary, which are not easily accessible. These hunting zones could then be assigned with higher costs or completely excluded from the cost surface.

6. Conclusion

In conclusion, the result of the cost distance analysis shows that most of the virtual boars live in the southern part of the Ruwer Valley. Hence, the vineyards in the centre of the study area are at high risk of being exploited by wild boars. In contrast, the northern part of the study area indicates high potential for the installation of vineyards, because only a few wild boars live there. Furthermore, these wild boars prefer to only forage the smaller vineyards near their initial location due to the shelter provided by the forest. However, the analysis overlooks the scenario where the vineyards do not provide sufficient food for the wild boars anymore. This could lead to the situation where the wild boars of the south travel further to the north and the wild boars in the northern area also exploit the larger vineyards further away from their initial location. Therefore, additional cost distance analyses are necessary. Moreover, the cost distance analysis could be improved by a lower weight of the roads, an optimization of the values of the different land cover classes and the implementation of further criteria such as hunting zones and soil types. The hypothesis that areas with abandoned vineyards are no longer in use due to increased risk of wild boar damage cannot be confirmed by the cost distance analysis. Although, the analysis suggests that wild boar damage can affect the use of vineyards, other environmental factors or a decreasing productivity, which were not investigated in this study, can also lead to the abandonment of vineyards.

7. References

- Brivio, F., Grignolio, S., Brogi, R., Benazzi, M., Bertolucci, C., & Apollonio, M. (2017). An analysis of intrinsic and extrinsic factors affecting the activity of a nocturnal species: The wild boar. *Mammalian Biology*, 84(1), 73–81.
<https://doi.org/10.1016/j.mambio.2017.01.007>
- Bruns, A. (2021). *No more Riesling?: Anpassung an den Klimawandel im Mosel-Weinbau* [Application/pdf]. <https://doi.org/10.25353/UBTR-XXXX-4A51-2203>
- Cappa, F., Lombardini, M., & Meriggi, A. (2019). Influence of seasonality, environmental and anthropic factors on crop damage by wild boar *Sus scrofa*. *Folia Zoologica*, 68(4), 261.
<https://doi.org/10.25225/fozo.015.2019>
- DELINAT. (n.d.). *Voraussetzungen und Standortansprüche | Weinwissen | Delinat*. Delinat - Wein aus reicher Natur. Retrieved 8 October 2024, from <https://www.delinat.com/weinberg-standort.html>
- geoportal RLP. (2024). *Web Map Service von Rheinland-Pfalz*.
<https://daten.rlp.de/geodata/87b2d177-c059-c04f-c780-d3ee540c5bde>
- German Wine Institut. (2024). *Region map*. <https://www.Deutscheweine.De/>.
<https://www.winesofgermany.com/our-regions/region-map>
- Grape to Glass. (2024). *Grape to Glass*. <https://grape-to-glass.com/index.php/flat-slope-vineyards/>
- IPBES. (2019). *Global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services*. Zenodo. <https://doi.org/10.5281/zenodo.6417333>
- Landesamt für Umwelt Rheinland-Pfalz. (2017). *Ruwer und Seitentäler | Fauna-Flora-Habitat-Gebiete in RLP*. Natura 2000 Bewirtschaftungspläne Und Steckbriefe.
https://natura2000.rlp.de/n2000-sb-bwp/steckbrief_gebiete.php?sbg_pk=FFH6306-301

- Li, M., Jiang, W., Li, B., & Butt, N. (2023). Social and cultural aspects of human–wildlife conflicts: Understanding people’s attitudes to crop-raiding animals and other wildlife in agricultural systems of the Tibetan Plateau. *Integrative Conservation*, 2(4), 214–225.
<https://doi.org/10.1002/inc3.30>
- Massei, G., Kindberg, J., Licoppe, A., Gačić, D., Šprem, N., Kamler, J., Baubet, E., Hohmann, U., Monaco, A., Ozoliņš, J., Cellina, S., Podgórski, T., Fonseca, C., Markov, N., Pokorný, B., Rosell, C., & Náhlik, A. (2015). Wild boar populations up, numbers of hunters down? A review of trends and implications for Europe. *Pest Management Science*, 71(4), 492–500.
<https://doi.org/10.1002/ps.3965>
- Ministerium für Wirtschaft Klimaschutz, Energie und Landesplanung Rheinland-pfalz, & Ministerium für Umwelt, Landwirtschaft Ernährung, Weinbau und Forsten Rheinland-pfalz. (2015). *Vineyard Soils of Rhineland-Palatinate: Rocks. Soils. Terroir*. Ministerium für Wirtschaft Klimaschutz, Energie und Landesplanung Rheinland-pfalz & Ministerium für Umwelt, Landwirtschaft Ernährung, Weinbau und Forsten Rheinland-pfalz.
https://www.lgb-rlp.de/fileadmin/service/lgb_downloads/boden/steinundwein/boden_des_jahres_2014_english.pdf
- Neuner, G., Huber, B., Plangger, A., Pohlin, J.-M., & Walde, J. (2020). Low temperatures at higher elevations require plants to exhibit increased freezing resistance throughout the summer months. *Environmental and Experimental Botany*, 169, 103882.
<https://doi.org/10.1016/j.envexpbot.2019.103882>
- Pittiglio, C., Khomenko, S., & Beltran-Alcrudo, D. (2018). Wild boar mapping using population-density statistics: From polygons to high resolution raster maps. *PLOS ONE*, 13(5), e0193295. <https://doi.org/10.1371/journal.pone.0193295>

- Richter, G. (1980). On the soil erosion problem in the temperate humid area of Central Europe. *GeoJournal*, 4(3), 279–287. <https://doi.org/10.1007/BF00218584>
- Rodrigo-Comino, J., Brings, C., Iserloh, T., Casper, M. C., Seeger, M., Senciales, J. M., Brevik, E. C., Ruiz-Sinoga, J. D., & Ries, J. B. (2017). Temporal changes in soil water erosion on sloping vineyards in the Ruwer- Mosel Valley. The impact of age and plantation works in young and old vines. *Journal of Hydrology and Hydromechanics*, 65(4), 402–409. <https://doi.org/10.1515/johh-2017-0022>
- Scillitani, L., Monaco, A., Toso, S. (2009). Do intensive drive hunts affect wild boar (*Sus scrofa*) spatial behaviour in Italy? Some evidences and management implications. *Eur J Wildl Res* (2010) 56:307–318 <https://doi.org/10.1007/s10344-009-0314-z>
- Schley, L., Dufrêne, M., Krier, A., & Frantz, A. C. (2008). Patterns of crop damage by wild boar (*Sus scrofa*) in Luxembourg over a 10-year period. *European Journal of Wildlife Research*, 54(4), 589–599. <https://doi.org/10.1007/s10344-008-0183-x>
- Woschek, H.-G. (2014). *Woscheks Weinwissen*. Woscheks-Weinwissen. <https://www.woscheks-weinwissen.de/doku.php/klima-boden-lage>

8. Appendix

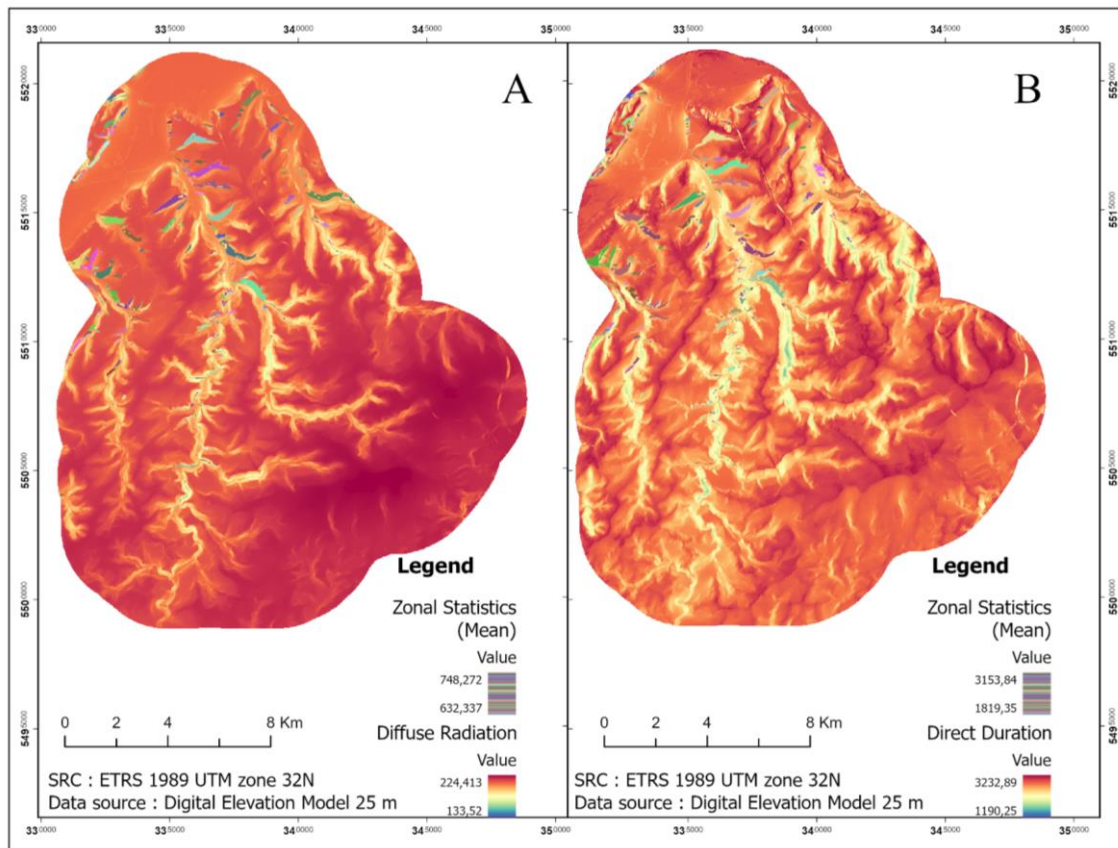


Figure 7: Solar Analysis result

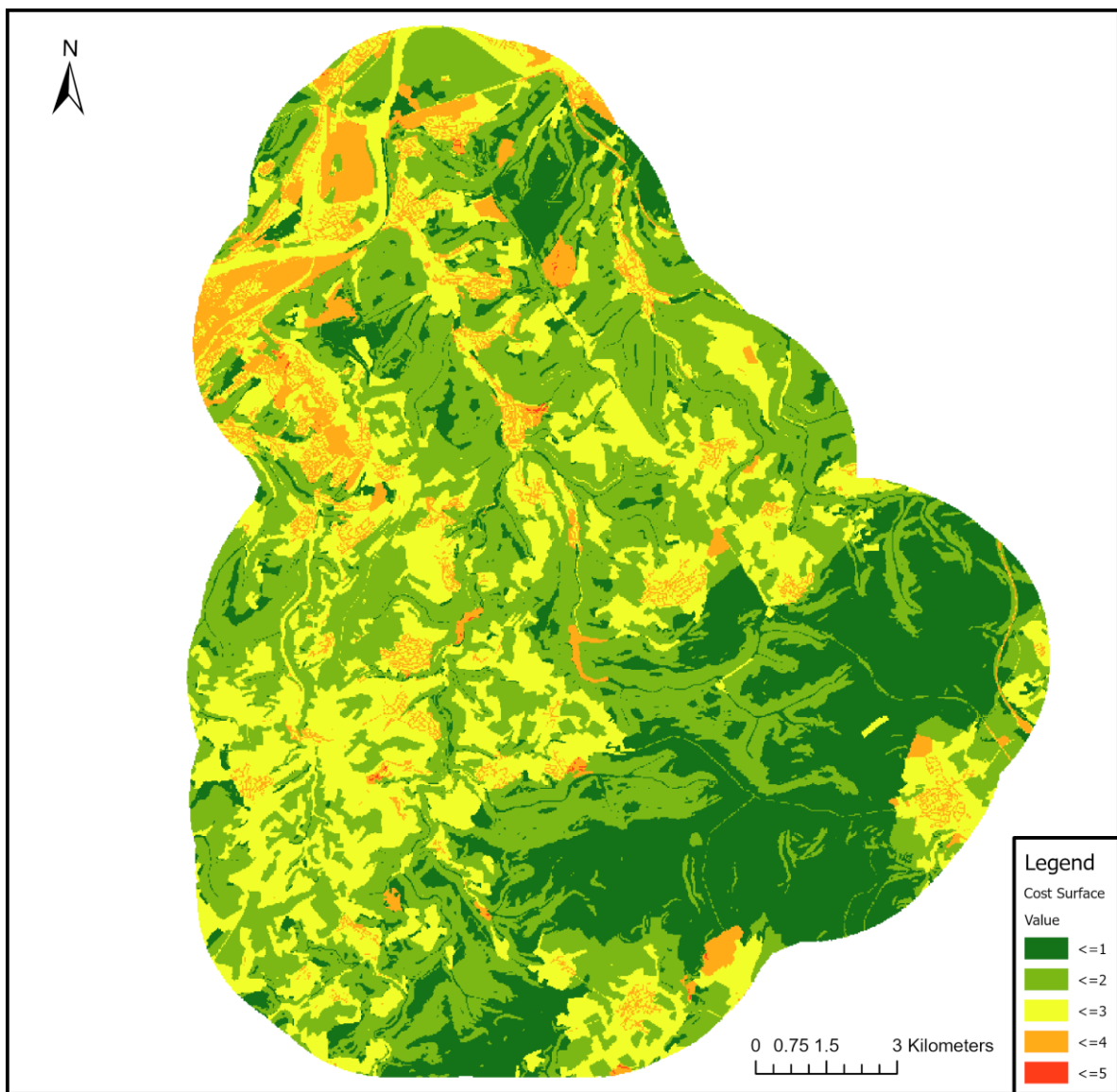


Figure 8: Cost Distance Surface